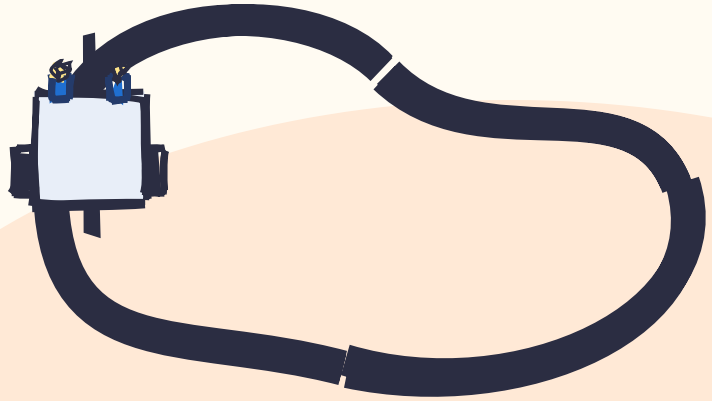
Junior Makers · Class 11

Line Following Robot

Your first robot that drives all by itself!

IR sensors | Motor driver | If / else decisions



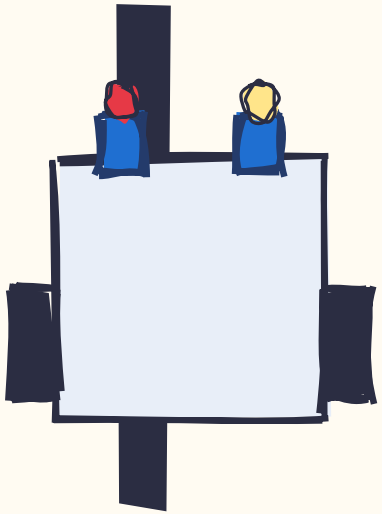
Did you know?
Hospital robots follow lines or maps to carry medicine and food along long corridors, day and night.

Let's think...

How does a robot know where to drive?

In a big warehouse

Thousands of parcels move every hour. Robots carry the shelves to the workers. Nobody is steering them. Some just follow **lines on the floor**. How can a robot 'see' a black line?



Which way now?

A camera?

Magic eyes?

Light that bounces!

Today: we build a robot with **two light sensors** that steers itself along a black line, and we race it!

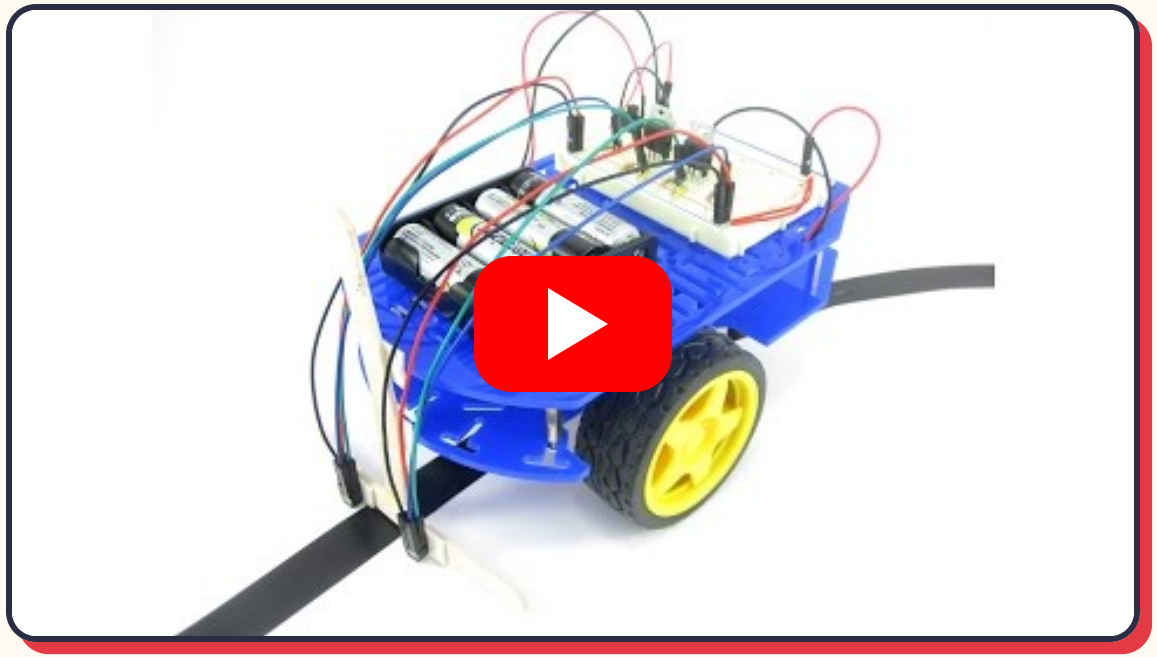
Learning objectives

By the end of this class I can...

1	**Sense:** explain how an **IR sensor** tells black from white	**Proof:** my sensor's 0 or 1 recorded on black and on white
2	**Steer:** make the robot turn by changing the speed of each **wheel**	**Proof:** my robot goes forward, turns left and turns right on command
3	**Decide:** write **if / else** rules that choose what the robot does	**Proof:** my code has a rule for all 4 sensor cases
4	**Tune:** test and **tune** the speed so the robot stays on the line	**Proof:** my robot finishes a full lap of the track

Objective 1 · Sense · video

Video time!



Line-Tracking Robot

Science Buddies · about 3 minutes · youtu.be/-cilOho6l3Q

Watch for...

1. What part of the robot **sees** the line?
2. What does the robot do when it starts to **drift** off the line?
3. Could this robot follow a **white** line on a black floor? How?

Scan to watch

After the video: Explain to your partner how the robot knows which way to turn. Use the words **sensor** and **wheel**.

Objective 1 · Sense

Black swallows light, white bounces it

The idea

The IR sensor has two 'eyes'. One **shines invisible infrared light** down. The other **catches** the light that bounces back.

White floor: lots of light bounces back → sensor says **0** (its red LED is on).
Black line: the light is swallowed → sensor says **1**.

Check yours

Some modules are the other way round (1 on white). That is why we always test first!

Example: black and white T-shirts

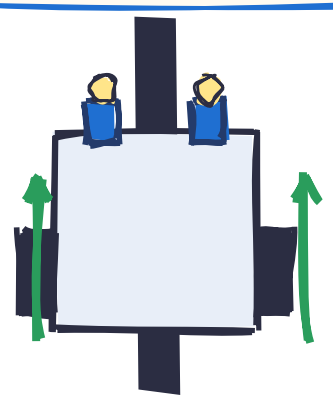


absorbs **reflects**

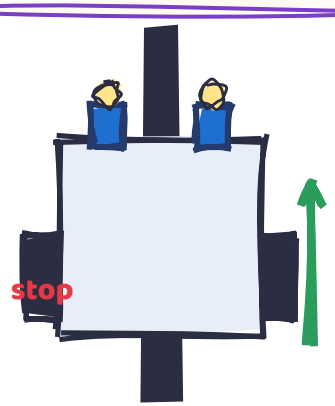
On a sunny day a **black** T-shirt feels hot: it **absorbs** the light. A **white** T-shirt stays cooler: it **reflects** the light away. Black tape does the same to the sensor's IR light.

Objective 2 · Steer

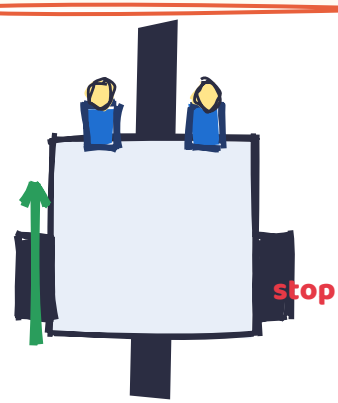
Steering with two wheels



Forward



Turn left



Turn right

Example: a wheelchair

Push only the **right** wheel of a wheelchair and it turns **left**. Push both equally and it goes straight. Tanks and bulldozers steer the same way.

The idea

Our robot has **no steering wheel**. To turn **left**, the **right** wheel keeps going and the **left** wheel stops (or slows). The robot swings towards the slower side.

Real world

Robots that follow paths



Warehouses
Robots carry shelves along floor lines and codes



Hospitals
Delivery robots carry medicine down corridors



Robot mowers
Follow a hidden wire round the garden



Cars
Lane-keeping cameras watch the road lines

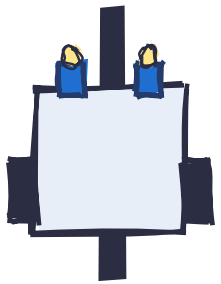


Where in our school could a line-following robot do a useful job?

Sensors + rules
Real robots use the same idea as ours, with more sensors and smarter rules.

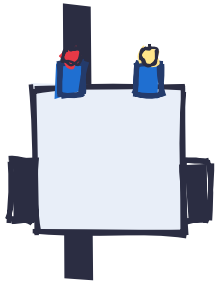
Objective 3 · Decide

Four cases, four rules



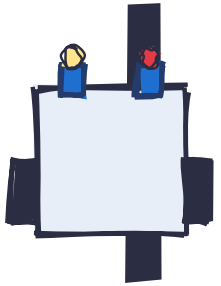
L = 0 · R = 0

Forward



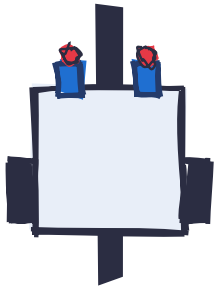
L = 1 · R = 0

Turn left



L = 0 · R = 1

Turn right



L = 1 · R = 1

Stop

Why left?

If the **left** sensor sees black, the line has moved to the left. Turn **towards** it!

Think: when would **both** sensors see black? (Hint: think of a finish line or a crossroads.)

Objective 4 · Tune

Tuning: fast or accurate?

The idea

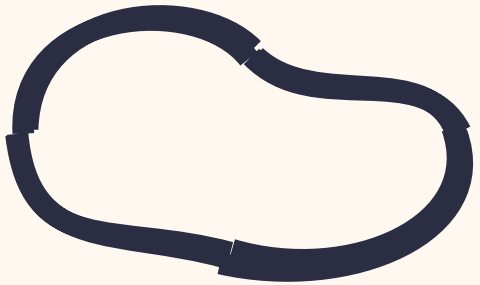
A **fast** robot reaches the bend before it can turn, and shoots off the line. A **slow** robot stays on the line but takes forever.

Engineers **tune**: change one number, test, write down what happened, change it again. Start with speed **150**.

Fair test

Change only ONE thing at a time, or you will not know what helped.

Example: riding a bicycle



slow on the bends!

You **slow down** before a sharp corner on a bike, then speed up on the straight. Our robot can learn the same trick: full speed when both sensors see white, slower when it turns.

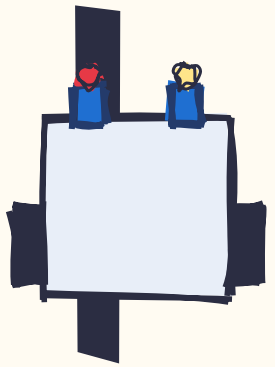
Quick quiz

What would the robot do?

Situation	Robot does…
1. Left = 0, Right = 0	_____
2. Left = 1, Right = 0	_____
3. Left = 0, Right = 1	_____
4. Left = 1, Right = 1	_____
5. Right wheel stops, left wheel spins	The robot turns _____

Answers

Cover this! 1 forward · 2 turn left · 3 turn right · 4 stop · 5 right



Part 2 · Make it

Now let's put it to work!

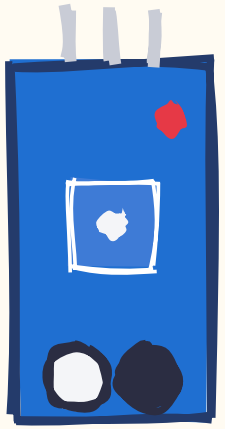
You know the ideas. Time to play, build and test. Use these slides while you work:

1. Sensor detective
2. Human line robot
3. Our robot kit (per group)
4. Build the chassis
5. Wire the motor driver
6. Wire the sensors and test the wheels
7. Code the rules in ArduinoBlocks

Objective 1 · Sense · test it

Sensor detective

Hold the sensor…	Value (0 or 1)	Red LED
Over white paper	_____	on / off
Over black tape	_____	on / off
Held up in the air	_____	on / off
Over your hand	_____	on / off



IR sensor

Blue screw

The little screw is the **sensitivity** knob. Turn it until the LED goes **off over black** and **on over white**.

How: wire VCC to 5V, GND to GND, OUT to pin 2. Upload a test that shows pin 2 on the Serial Monitor. Hold the sensor **1 cm** above each surface.

Prediction first!
Before you test, guess every answer. Were you right?

Class game

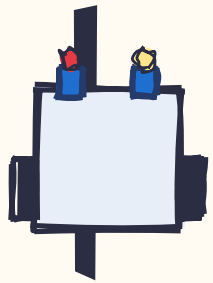
Human line robot

How to play

1. Tape a curvy black line on the floor (or use a skipping rope).
2. One student is the **robot** and walks slowly, eyes on the ceiling!
3. Two 'sensor' students walk at the robot's front left and front right, watching the line.
4. A sensor that sees the line calls its rule: **"Left sensor: turn left!"**
5. No sensor sees it? **"Straight!"** Both see it? **"Stop!"**

Then discuss

What happens if the robot walks too **fast**? What if the sensors are too **far apart**?



Big idea

The robot does not understand the line. It just follows simple rules very fast, many times every second.

Build it

Our robot kit (per group)



Arduino Uno
The brain + USB cable



2WD chassis
2 motors, 2 wheels, castor



2 × IR sensor
The robot's eyes



L298N driver
Gives motors power



4 × AA pack
Motor power



Screwdriver
For the chassis

Also: jumper wires, black electrical tape and white chart paper for the track, your JM-10 sensor mount if it was printed.

Batteries
Use fresh batteries. Weak batteries make the robot slow and wobbly.

Objective 2 · Build step 1 of 4

Build the chassis

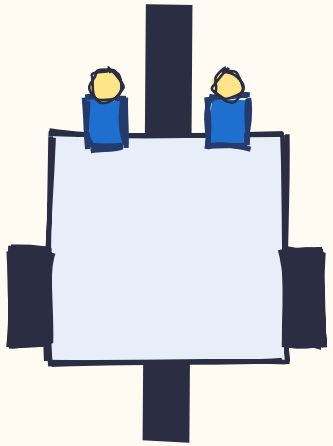
a Screw the two **motors** under the chassis plate. Push the **wheels** on.

b Fix the **castor wheel** at the front so the robot rolls smoothly.

c Mount the **Arduino** and the **L298N** on top with screws or double-sided tape.

d Fix the two **IR sensors** at the front, pointing down, about **1 cm** above the floor and **2 to 3 cm** apart.

e The black line must fit **between** the two sensors.



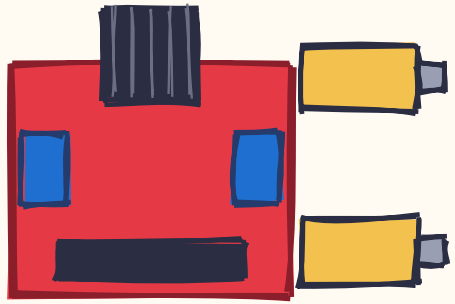
Check: Wheels spin freely and the sensors sit just above the floor on either side of the line.

Do not overtighten the screws: the plastic plate can crack.

Objective 2 · Steer · Build step 2 of 4

Wire the motor driver

From	To
Battery + / -	L298N 12V / GND
L298N GND	Arduino GND (shared ground!)
Left motor	OUT1 + OUT2
Right motor	OUT3 + OUT4
ENA / ENB	pins 5 / 6
IN1 / IN2	pins 7 / 8
IN3 / IN4	pins 9 / 10



L298N driver

Remove the caps

Pull off the little black **jumper caps** on ENA and ENB so we can control **speed**.

Shared ground

The Arduino and the L298N **must** share GND, or the motors ignore the Arduino.

Power

Arduino runs from the USB cable or a power bank. Motors run from the AA pack.

Objective 2 · Build step 3 of 4

Wire the sensors and test the wheels

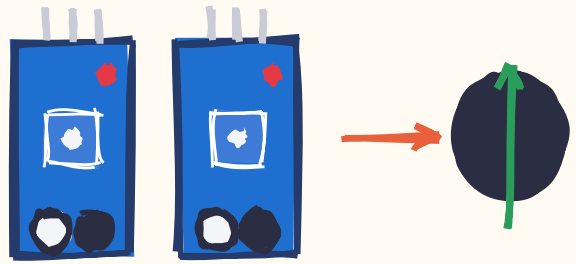
a Left IR: VCC to 5V, GND to GND, OUT to **pin 2**. Right IR: OUT to **pin 3**.

b Add the **LED** on pin 12 (with 220 Ω) and the **buzzer** on pin 11.

c Upload the **wheel test**: both wheels forward for 2 seconds.

d A wheel spinning backwards? **Swap its two wires** on the L298N.

e Lift the robot off the table while testing!



Check: Both wheels spin forward and each sensor's LED changes over black tape.

Objective 3 · Decide · Build step 4 of 4

Code the rules in ArduinoBlocks

```
when Arduino starts
    set speed to 150
forever
    set L to digital read pin 2 · set R to digital read pin 3
    if L = 0 and R = 0 → forward (both wheels at speed)
    else if L = 1 and R = 0 → turn left (left wheel stop, right wheel speed)
    else if L = 0 and R = 1 → turn right (left wheel speed, right wheel stop)
    else → stop both wheels · LED on · beep
```

Forward means

IN1 HIGH, **IN2** LOW (left)
IN3 HIGH, **IN4** LOW (right)
ENA / **ENB** = speed (0 to 255)

Use functions
Make blocks called forward, left, right and stop. Your loop then reads like a story!

Objective 4 · Tune · test and tune

Lap time challenge

Try	Speed	Lap time (s)	Stayed on? (Y/N)	What we noticed
1	150			
2	180			
3	120			
4	your best			

How to time

One student starts the stopwatch when the robot crosses the start tape and stops it after **one full lap**.

Class record

Fastest lap that stays on the line goes on the board!

Debug: spins in circles? A sensor or motor is swapped. Ignores the line? Turn the blue screw. Stops everywhere? The sensors are too far from the floor.

MAKE

Level up!

Robot challenges

Choose your challenge

Smooth turns: slow the inside wheel instead of stopping it (speed 60).

Finish line: when both sensors see black, stop and play a victory tune.

White line: change the rules so the robot follows a white line on black card.

Mount it: fit the JM-10 3D printed sensor mount and test again.

Think like an engineer

Why do real delivery robots use more than two sensors?

Boss level

Add a third sensor in the middle. Write rules for all 8 cases!

Objective check

Can I really do it? Prove it.

01	Explain why the sensor gives a different number on black and on white.	○ ○ ○
02	Show your robot turning left. Which wheel is moving and which is stopped?	○ ○ ○
03	Point to the rule in your code for **L = 1, R = 0**. What does the robot do?	○ ○ ○
04	Show your lap time table. Which speed worked best, and why?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Run a class race at the end: only robots that finish a lap are timed.

Show what you know

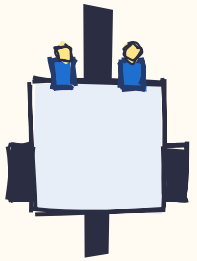
What did we learn?

- [x] Black absorbs IR light, white reflects it
- [x] The IR sensor turns that into 0 or 1
- [x] Turn by stopping or slowing one wheel
- [x] Four if / else rules steer the robot
- [x] Engineers tune one number at a time

Exit ticket: the left sensor sees black. Which way should the robot turn, and why?

At home

Find 3 machines at home that use sensors to decide what to do.



Next class: build a Phone-Controlled RGB Mood Lamp!